2025 ANNUAL REPORT *EARTH SCIENCE TECHNOLOGY OFFICE*

National Aeronautics and Space Administration

About ESTO

As the technology development function within NASA's Earth Science Division, the Earth Science Technology Office (ESTO) performs strategic planning and manages the development of a broad range of nascent technologies for future science measurements. ESTO relies on competition and peer review to select the best cutting-edge technologies, from advanced sensors aboard miniature satellites to software tools that plan new observations and harmonize, fuse, and analyze large data sets from various sources.

Our approach to Technology Development:

- **Strategy:** Engage with the Earth science community to plan investments through careful analyses of science requirements
- **Selection:** Fund technology development through periodic, competitive solicitations and partnership opportunities
- **Management:** Review and advise funded technology projects on progress and performance
- **Infusion:** Encourage and facilitate the use of matured technologies in science measurements

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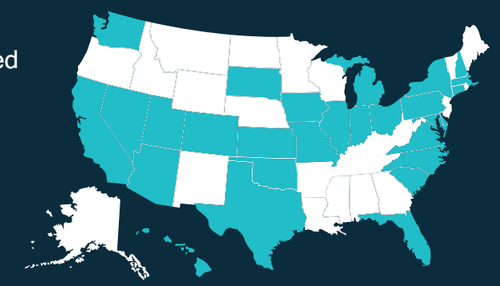
By The Numbers FY25

58 Projects Added in FY25

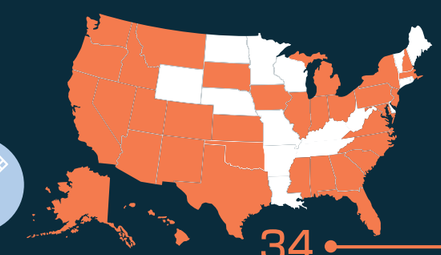
184 Projects Active in FY25

60 Projects Graduated in FY25



71 Unique PI Orgs

28 States



162 Unique Co-I Orgs

282 Students

34 States

3 Technology Demonstrations On Orbit

74 Colleges & Universities

11 Projects Airborne Tested

11 Journal Articles Published

370 Conference Publications

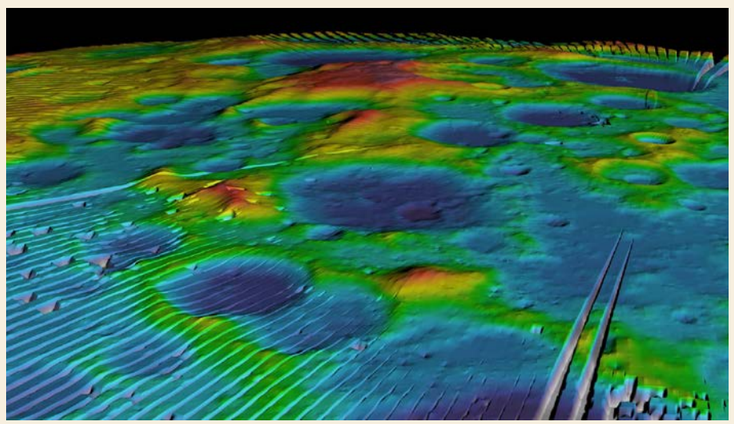
1 Patents

METRICS FY25

GOAL 3: SCIENCE MEASUREMENT

>> Enable a new science measurement or significantly improve the performance of an existing technique. Several projects satisfied this goal in FY25. One notable example follows:

Project Highlight: An AI-Powered Lidar for Mapping Earth's Surface



3D topography data describing the Moon's surface. CASALS could help scientists select potential landing sites for future lunar missions. Credit: NASA

Engineer Jeffrey Chen tests a lidar prototype on the roof of Building 33 at NASA's Goddard Space Flight Center in Greenbelt, Maryland. Chen and his team earned the center's 2024 Innovator of the Year award for their work on CASALS, a lidar system enhanced with artificial intelligence and other technologies. Credit: NASA

The Concurrent Artificially-intelligent Spectrometry and Adaptive Lidar System (CASALS), a novel lidar instrument developed by researchers at NASA's Goddard Space Flight Center, flew aboard a Beechcraft B200 Super King Air in October and November of 2024, marking a milestone in the quest for improved space-based altimeters.

CASALS adaptively distributes its laser beam to specified locations across a swath using a photonic integrated circuit seed laser, wavelength tuning circuitry, a high-power fiber amplifier, and a wavelength-to-angle mapping dispersive grating. This capability not only ensures measurement continuity with NASA's ICESat-2 and GEDI missions, but also, for the first time from space, images the observed swath in 3D. This will allow researchers to map Earth's surface and planetary boundary layer with unprecedented precision, improving ecosystem models and forecasts of fresh water supplies. Researchers could also use CASALS to produce detailed maps of the moon, giving scientists a better look at the subsurface structures beneath potential lunar landing sites.

CASALS integrates adaptive lidar technologies, hyperspectral imaging, and onboard artificial intelligence within a single instrument to reduce cost and improve performance when compared to traditional lidars, and its compact size makes it ideal for use aboard small satellites. Between October 30 and November 29, 2024, CASALS completed critical flight tests out of NASA's Langley Research Center (LaRC). Now, the team is eyeing future opportunities to fly their instrument in space.

PROGRAM UPDATES FY25

Each of our program elements addresses a different aspect of Earth science technology, from modelling and information systems to full instruments and instrument subsystems. The following section details some of the latest developments in each of these programs and highlights a few projects that achieved significant milestones, including work dedicated to Earth System Digital Twins, quantum sensing, and metamaterials.

In-space Validation of Earth Science Technologies

NASA's vision for future Earth observations necessitates the development of emerging technologies capable of making new or improved Earth science measurements. Promising new capabilities, however, bring complexity and risk, and for some technologies there remains a critical need for validation in the hazardous environment of space. ESTO's In-space Validation of Earth Science Technologies (InVEST) program facilitates the space demonstration of technology projects that cannot be sufficiently evaluated on the ground or through airborne testing. Once validated in space, technologies are generally more adoptable, even beyond their intended use.

11 Projects Active in FY25

1 Project Added in FY25

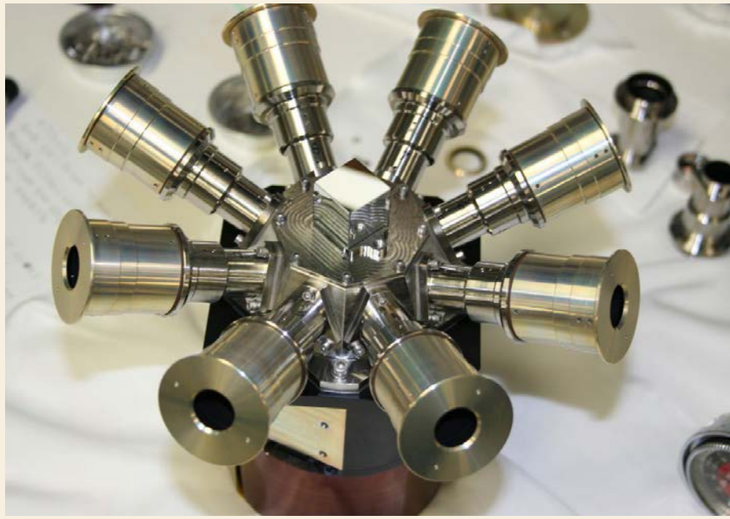
AFFORD - Advanced Format and Feature Optimized ROIC Design (component technology for future flight) — Atul Joshi, Saaz Micro Inc

1 Project Graduated

Compact Total Irradiance Monitor Flight Demonstration — David Harber, University of Colorado Boulder

InVEST Project Highlight

On March 15, the Aerosol Radiometer for Global Observation of the Stratosphere (ARGOS) InVEST technology demonstration mission launched aboard the SpaceX Transporter 13 Smallsat Rideshare vehicle. ARGOS will demonstrate the simultaneous collection of limb scattering data from eight viewing directions, a novel approach that has the potential to improve aerosol sensitivity globally when compared to current instruments.



A photo of ARGOS's eight-track telescope, which will help researchers characterize aerosols more accurately. Credit: M. DeLand / Science Systems and Applications, Inc

SENSING WILDFIRE

A novel space-based sensor for observing wildfires could allow first responders to monitor burns at a global scale, paving the way for future small satellite (SmallSat) constellations dedicated entirely to fire management and prevention.

Developed with ESTO's support, the "Compact Fire Infrared Radiance Spectral Tracker" (c-FIRST) is a small, mid-wave infrared sensor that collects thermal radiation data across five spectral bands. Most traditional space-based sensors dedicated to observing fires have long revisit times, observing a scene just once over days or even weeks. The compact c-FIRST sensor could be employed in a SmallSat constellation that could observe a scene multiple times a day, providing first responders data with high spatial resolution in under an hour.

In addition, c-FIRST's dynamic spectral range covers the entire temperature profile of terrestrial wild fires, making it easier for first responders to detect everything from smoldering, low-intensity fires to flaming, high intensity fires.

"Wildfires are becoming more frequent, and not only in California. It's a worldwide problem, and it generates tons of by-products that create very unhealthy conditions for humans," said Sarath Gunapala, who is an Engineering Fellow at NASA's Jet Propulsion Laboratory (JPL) and serves as Principal Investigator for c-FIRST.

The need for space-based assets dedicated to wildfire management is severe. During the Palisade and Eaton Fires earlier this year, strong winds kept critical observation aircraft from taking to the skies, making it difficult for firefighters to monitor and track massive burns.

Space-based sensors with high revisit rates and high spatial resolution would give firefighters and first responders a constant source of eye-in-the-sky data.

"Ground-based assets don't have far-away vision. They can only see a local area. And airborne assets, they can't fly all the time. A small constellation of CubeSats could give you that constant coverage," said Gunapala.

c-FIRST leverages decades of sensor development at JPL to achieve its compact size and high performance. In particular, the quarter-sized High Operating Temperature Barrier Infrared Detector (HOT-BIRD), a compact infrared detector also developed at JPL with ESTO support, keeps c-FIRST small, eliminating the need for bulky cryocooler subsystems that add mass to traditional infrared sensors.

With HOT-BIRD alone, c-FIRST gathers high-resolution images and quantitative retrievals of targets between 300°K (about 80°F) to 1000°K (about 1300°F). But when paired with a state-of-the-art Digital Readout Integrated Circuit (DROIC), c-FIRST can observe targets greater than 1600°K (about 2400°F).

Developed by Copious Imaging LLC and JPL with funding from NASA's Small Business Innovation Research (SBIR) program, this DROIC features an in-pixel digital counter to reduce saturation, allowing c-FIRST to capture reliable infrared data across a broader spectral range.

Artificial intelligence (AI) will also play a role in c-FIRST's success. Gunapala plans to leverage AI in an onboard smart controller that parses collected data for evidence of hot spots or active burns. This data will be prioritized for downlinking, keeping first responders one step ahead of potential wildfires.

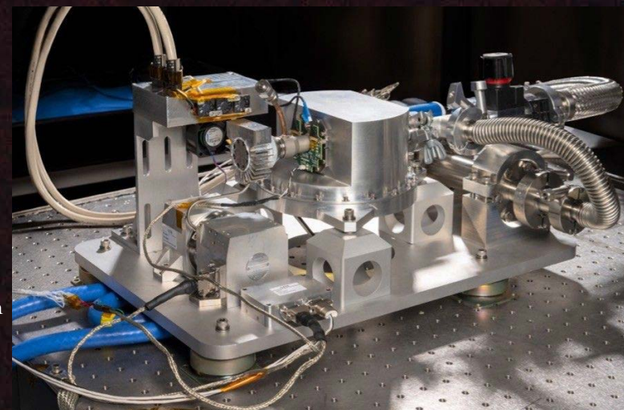
"We wanted it to be simple, small, low cost, low power, low weight, and low volume, so that it's ideal for a small satellite constellation," said Gunapala.

Gunapala and his team had a unique opportunity to test c-FIRST after the Palisade and Eaton Fires in California. Flying their instrument aboard NASA's B-200 Super King Air, the scientists identified lingering hot spots in the Palisades and Eaton Canyon area five days after the initial burn had been contained.

Now, the team is eyeing a path to low Earth orbit. Gunapala explained that their current prototype employs a standard desktop computer that isn't suited for the rigors of space, and they're working to incorporate a radiation-tolerant computer into their instrument design.

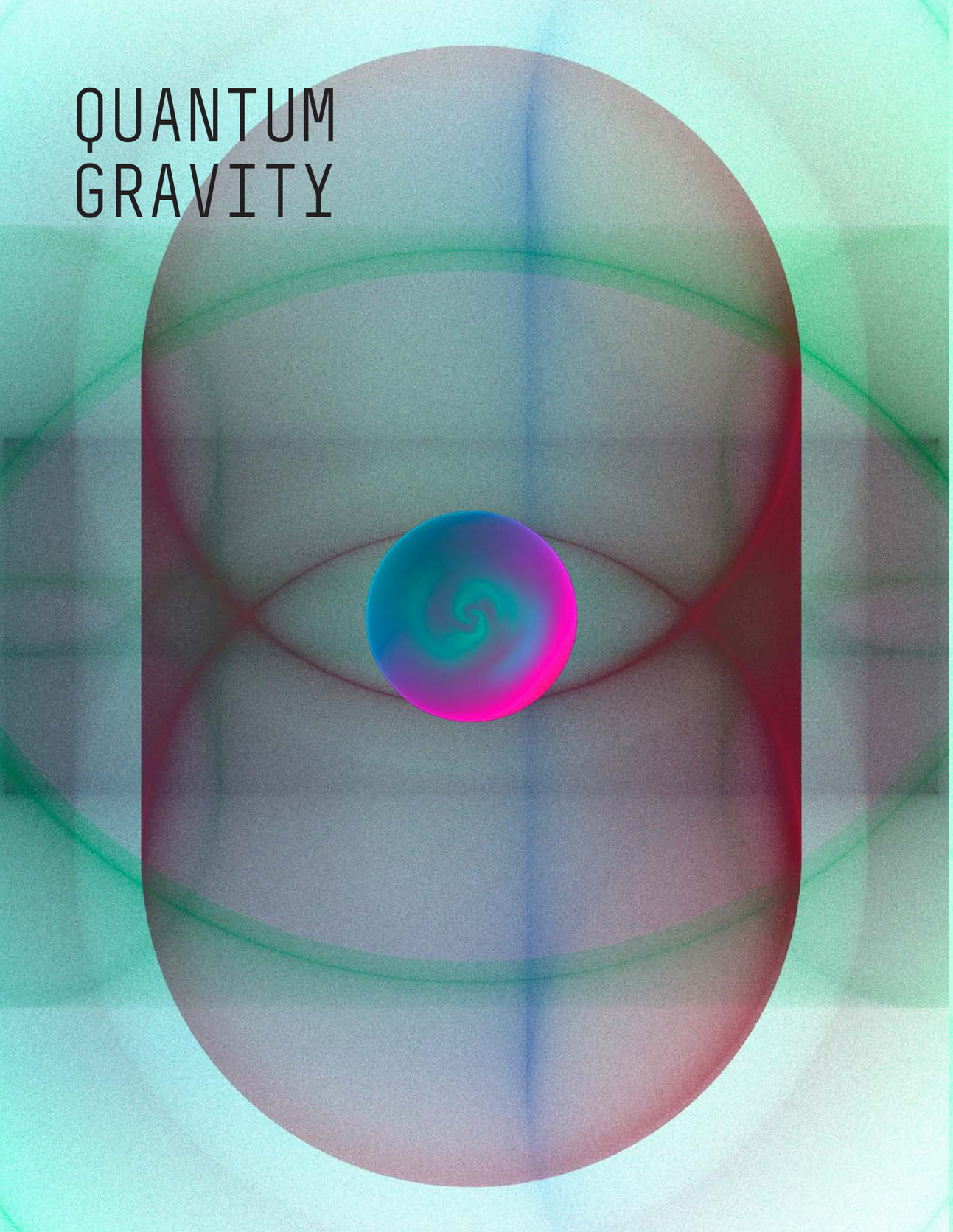
But this successful test over Los Angeles demonstrates c-FIRST is fit for fire detection and science applications. As wildfires become increasingly common and more destructive, Gunapala hopes that this tool will help first responders combat nascent wildfires before they become catastrophes.

"To fight these things, you need to detect them when they're very small," said Gunapala.



˅ NASA's c-FIRST instrument could provide high resolution data from a compact space-based platform in under an hour, making it easier for wildfire managers to detect and monitor active burns. Credit: NASA

QUANTUM GRAVITY



Researchers from NASA's Jet Propulsion Laboratory in Southern California, private companies, and academic institutions are developing the first space-based quantum sensor for measuring gravity. Supported by ESTO, this mission will mark a first for quantum sensing and will pave the way for groundbreaking observations of everything from petroleum reserves to global supplies of fresh water.

Earth's gravitational field is dynamic, changing each day as geologic processes redistribute mass across our planet's surface. The greater the mass, the greater the gravity.

You wouldn't notice these subtle changes in gravity as you go about your day, but with sensitive tools called gravity gradiometers, scientists can map the nuances of Earth's gravitational field and correlate them to subterranean features like aquifers and mineral deposits. These gravity maps are essential for navigation, resource management, and national security.

"We could determine the mass of the Himalayas using atoms," said Jason Hyon, chief technologist for Earth Science at JPL and director of JPL's Quantum Space Innovation Center. Hyon and colleagues laid out the concepts behind their Quantum Gravity Gradiometer Pathfinder (QGGPf) instrument in a recent paper in EPJ Quantum Technology.

Gravity gradiometers track how fast an object in one location falls compared to an object falling just a short distance away. The difference in acceleration between these two free-falling objects, also known as test masses, corresponds to differences in gravitational strength. Test masses fall faster where gravity is stronger.

QGGPf will use two clouds of ultra-cold rubidium atoms as test masses. Cooled to a temperature near absolute zero, the particles in these clouds behave like waves. The quantum gravity gradiometer will measure the difference in acceleration between these matter waves to locate gravitational anomalies.

Using clouds of ultra-cold atoms as test masses is ideal for ensuring that space-based gravity measurements remain accurate over long periods of time, explained Sheng-wey Chiow, an experimental physicist at JPL. "With atoms, I can guarantee that every measurement will be the same. We are less sensitive to environmental effects."

Using atoms as test masses also makes it possible to measure gravity with a compact instrument aboard a single spacecraft. QGGPf will be around 0.25 cubic meters (0.3 cubic yards) in volume and weigh only about 125 kilograms (275 pounds), smaller and lighter than traditional space-based gravity instruments.

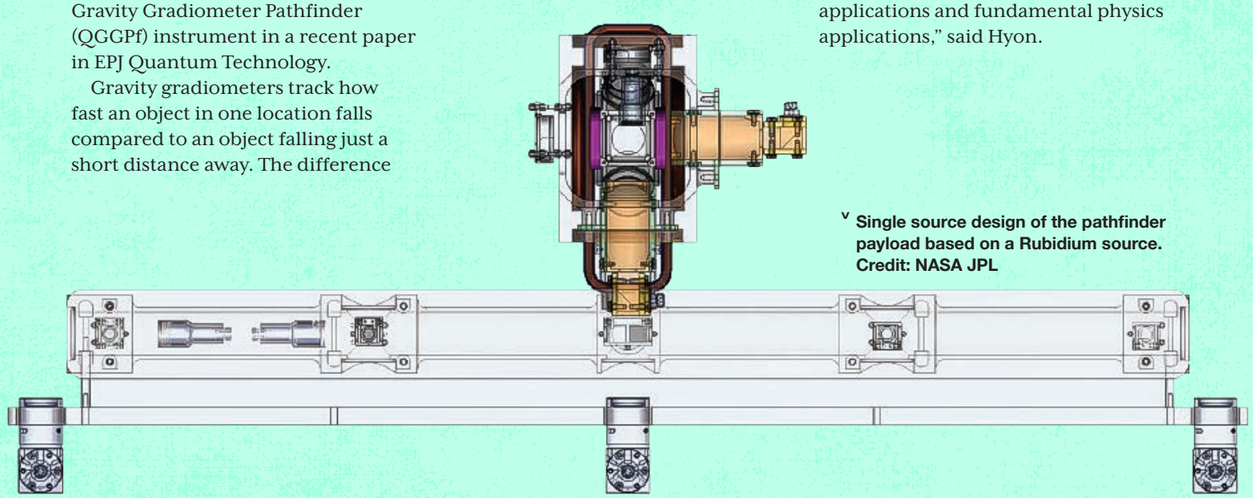
Quantum sensors also have the potential for increased sensitivity. By some estimates, a science-grade quantum gravity gradiometer instrument could be as much as ten times more sensitive at measuring gravity than classical sensors.

The main purpose of this technology validation mission, scheduled to launch near the end of the decade, will be to test a collection of novel technologies for manipulating interactions between light and matter at the atomic scale.

"No one has tried to fly one of these instruments yet," said Ben Stray, a postdoctoral researcher at JPL. "We need to fly it so that we can figure out how well it will operate, and that will allow us to not only advance the quantum gravity gradiometer, but also quantum technology in general."

This technology development project involves significant collaborations with small businesses to advance the sensor head technology and the laser optical system.

Ultimately, the innovations achieved during this pathfinder mission could enhance our ability to study Earth, and our ability to understand distant planets and the role gravity plays in shaping the cosmos. "The QGGPf instrument will lead to planetary science applications and fundamental physics applications," said Hyon.



Single source design of the pathfinder payload based on a Rubidium source. Credit: NASA JPL

UNFOLDING NEW SCIENCE

A team of researchers at NASA's Goddard Space Flight Center are taking cues from the ancient Japanese art of paper folding to develop a compact deployable telescope, which could dramatically reduce the cost of sending lidar missions into space.

The "Metalens Origami Deployable Lidar Telescope (MODeL-T)," developed with support from ESTO, would be a composite lens made of more than 50 segments that fold together to form a cooler-sized cube. In space, that compact cube would unfold into a flat, star-shaped structure nearly two meters wide, tailored to shuttle direct captured lidar signals towards a receiver. Compact at launch and large in space, MODeL-T could allow small space craft to host lidar instruments as powerful as those found on larger, more expensive platforms.

Compact lidar systems equipped with MODeL-T would fit aboard ESPA-class satellites, allowing scientists to take advantage of cost-efficient rideshare launch opportunities. Compared to previous missions, lidar equipped with MODeL-T could reduce the cost of space-based lidar by as much as a factor of 20.

"It's like moving from launching a large refrigerator into orbit to launching a suitcase," said Mark Stephen, a Research Engineer at GSFC and Principal Investigator for MODeL-T.

Stephen, who helped develop the lidar instruments for ICESat and ICESat-2, explained that MODeL-T would replace the large receiver telescopes traditional lidar instruments rely on to capture photons and guide them towards a detector, which are essentially empty tubes with large beryllium optics.

MODeL-T would be smaller than these traditional telescopes at launch, but larger once deployed. Whereas the telescope aboard ICESat-2 has an aperture one meter wide and takes up 400 liters of stowed space, MODeL-T would have an aperture 1.8 meters wide and take up just 100 liters of stowed space.

Larger apertures mean more photons reach a lidar detector, which improves data resolution and lowers the power requirements for lidar transmitters. If fewer photons are lost more return photons are captured per laser pulse, a lidar instrument can employ a smaller, lighter laser without sacrificing an effective signal-to-noise ratio.

To develop MODeL-T, Stephen's team overcame two significant technology hurdles: first, finding a flat, foldable material that manipulates light with the same reliability as traditional bulk optics; second, creating a lightweight mechanism for unfolding that flat material once in space.

Xingjie Ni, Associate Professor of Electrical Engineering at Pennsylvania State University, led the development of the foldable metamaterial optic, while Larry Howell, Associate Academic Vice President and Professor at Brigham Young University, led the development of the deployment mechanism.

Metamaterials use complex arrays of sub-micrometer-sized structures arranged across a flat surface to manipulate light. Ni, a metamaterials expert, said producing a metamaterial split across more than 50 unique elements that operate as a single, unified aperture was a rewarding challenge.

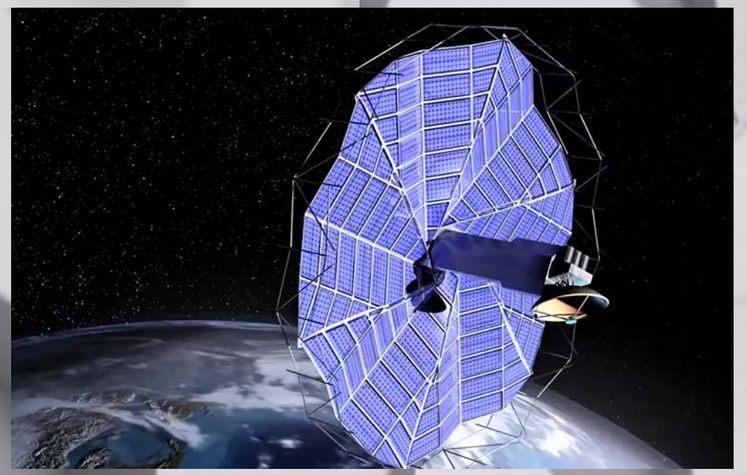
"I think that's why we are very excited about this. We can make this metamaterial big enough that it can be used for space applications, in a way that no one else has done before," Ni said.

Howell, who specializes in deployment mechanisms, said that the toughest challenge of this particular project has been ensuring the unfolded metamaterial segments are deploy perfectly aligned.

"We've done deployable solar panels, for example, and if you're off by a degree or two, it doesn't really impact performance. Here, the margin for error is much smaller," Howell said.

Each of these key components–the metamaterial optic and the deployment mechanism–have been demonstrated in a laboratory environment. Now, Stephen and his team are working to incorporate them within a single prototype.

"Lowering the cost of space-based lidar, turning this into an operational tool where we're getting more science data, that's really what I hope to achieve out of it all," said Stephen.



˅ A illustration of a future MoDEL-T instrument Credit: Ux Connections

Tech Demos On Orbit

ESTO validates new technologies in the harsh environment of space with the aim to reduce the overall risk of incorporating new technologies into future Earth science missions. Recent advancements in small, standardized satellites and low-cost access to space have transformed the way technologies are demonstrated and validated. Meaningful risk reduction for hardware components and information systems can be accomplished quickly for future flight instruments. Here are some recent highlights:

Launched in FY25

ARGOS

The Aerosol Radiometer for Global Observation of the Stratosphere (ARGOS), launched March 14, 2025, is a hosted payload instrument that is testing a new technique for observing atmospheric aerosols. Leveraging eight distinct limb views, the compact instrument will be useful for understanding how wildfires, volcanoes, and severe weather events might affect aerosols.

Arcstone

Arcstone, launched June 23, 2025, is testing a new on-orbit calibration technique using the Moon as a reference to ensure space-based Earth science instruments remain accurate over time. From a 6-unit CubeSat, the hyperspectral instrument will provide lunar spectral reflectance measurements needed to establish an absolute lunar calibration standard for past, current, and future Earth observing sensors.

Dynamic Targeting

Dynamic Targeting is a new AI-driven capability that enables satellites to analyze imagery onboard and autonomously decide where to point their instruments. Demonstrated on the CogniSAT-6 CubeSat, the system can look ahead along the spacecraft's orbit, detect clouds in real time, and capture only clear-sky imagery—all in under 90 seconds. This approach increases the amount of usable data returned from orbit and paves the way for future autonomous targeting of short-lived events such as storms, wildfires, and volcanic eruptions.

Launching in FY26

SUM Component

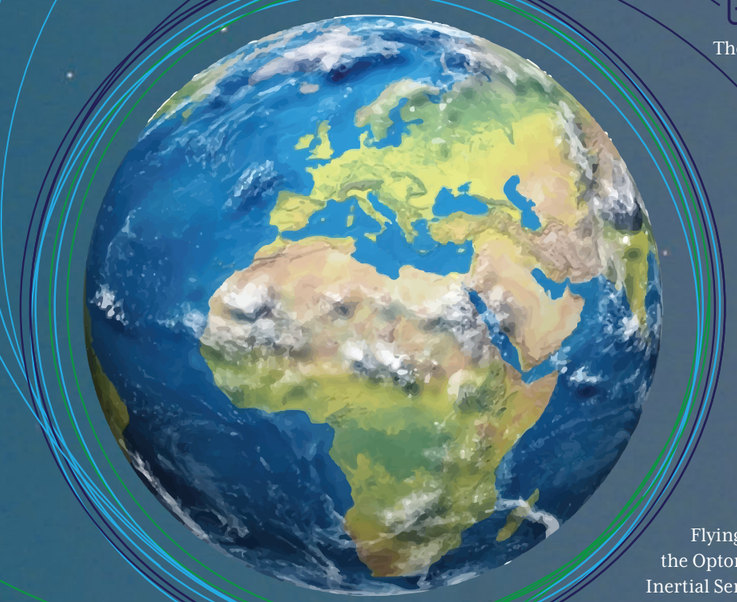
A specialized joint will be demonstrated in space for a Solid Underconstrained Multi-Frequency (SUM) deployable antenna. The overall project is developing a large (2.4 m), multi-segment, parabolic aperture that can deploy from a compact volume, enabling low-cost constellations of radars and radiometers for a variety of Earth and planetary observations.

GRITTS

The Geodetic Reference Instrument Transponder for Small Satellites (GRITTS) will verify a new geodetic measurement concept on a 16-unit CubeSat. Functioning as a GNSS L-to-S-and-X-band transponder, GRITTS could pave the way for a more cost-effective mission, employing a constellation of spacecraft for better global coverage and improved accuracy.

ACMES

Also a 16-unit CubeSat, the Active Cooling for Multispectral Sensor (ACMES) instrument will prototype a complete end-to-end solution for active thermal control of cryogenic instruments on nano- and small-satellites, using the previously developed Hyperspectral Thermal Imager (HyTI) as its test instrument. ACMES will also fly two student projects: the Filter Incidence Narrow-band Infrared Spectrometer (FINIS); and the Planer Langmuir/Impedance Diagnostic (PLAID).



Launching in FY27

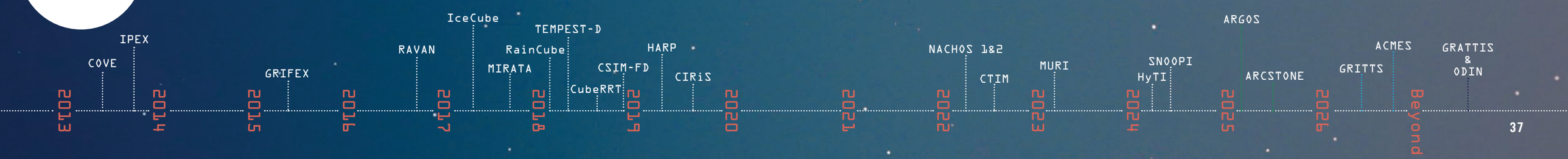
GRATTIS

The Gravitational Reference Advanced Technology Test In Space (GRATTIS) mission will flight-test the Simplified Gravitational Reference Sensor (S-GRS), a precision inertial sensor essential for future Earth geodesy missions. By validating S-GRS performance in orbit, GRATTIS will enable the use of a domestically produced, higher-precision sensor designed for drag-compensated operations.

ODIN

Flying as a secondary payload with GRATTIS, the Optomechanical Distributed Instrument for Inertial Sensing and Navigation (ODIN) will deliver a compact, ultra-sensitive inertial sensing system tailored for future mass-change missions. High-precision accelerometry is essential for detecting subtle shifts in Earth's gravity field, and ODIN offers a low-cost, low-SWaP approach to meeting this need.

Timeline of ESTO Launches



STUDENT SPOTLIGHT FY25

Students are integral to the work and success of technology development teams. Since ESTO's founding, at least 1,415 students from 186 institutions have worked on various ESTO-funded projects. Aided by their experiences, these students have often gone on to work in the aerospace industry and in related fields. In FY25 alone, at least 282 students from 74 institutions were involved with active technology development projects.

Phoebe Pan

Phoebe Pan is a freshman at Harvard University, where she plans to study Computer Science and Human Developmental and Regenerative Biology. Supporting Chaowei Yang's AIST project, Pan helped create a Generative Adversarial Network (GAN) for missing aerosol optical depth data manipulation and managed an independent research project that investigated new techniques for predicting PM2.5 pollution emitted by wildfires. "Working on this project helped me realize how much I enjoy applying computational techniques to solve real-world problems," Pan said. Pan is also an avid artist and crafter, and she recently started playing the steelpan.

Humphrey Chen

Humphrey Chen recently graduated from the University of California, Davis, with his PhD in Electrical Engineering. Supporting Ben Yoo's IIP project, Chen helped design and fabricate Photonic Integrated Circuits (PICs). He also worked on measurement and assembly test beds for the PIC designs. "The experience I gained from the project has greatly enhanced my skills, allowing me to quickly measure PIC-based systems and work towards higher TRL scientific instruments," said Chen. Chen's hobbies include art, especially ceramics, and weightlifting.

Aashish Panta

Aashish Panta is a fifth year PhD student at the University of Utah. As a member of Kyo Lee's AIST project, Panta leveraged his expertise in artificial intelligence to help build integrated retrieval-augmented generation pipelines to help process large Earth science data sets more efficiently. "These experiences, as a whole, helped me develop the ability to design scalable AI-driven systems for scientific data, work collaboratively across disciplines and turn research into impactful solutions," Panta said. In his free time, Panta enjoys soccer, hiking, skiing, and traveling.

Leila Freitag

Leila Freitag is a third-year graduate student at MIT, where she studies mechanical engineering. Supporting John Leckey's SPECIES project, which aims to create deployable diffractive optics for low-SWaP, space-based lidar, Freitag developed a custom model to assess system design tradeoffs and choices. The model uses signal-to-noise ratio as a performance metric, and it will be used to help inform feasible mission architectures. Freitag is also analyzing ground-based lidar data gathered by teammates at NASA's Langley Research Center. "I really value John's mentorship and career advice," she said, adding that she'd like to leverage her experience with NASA to continue researching exciting problems after graduation. Outside of the lab, Freitag enjoys backpacking and making pottery.

Jaiden Luna

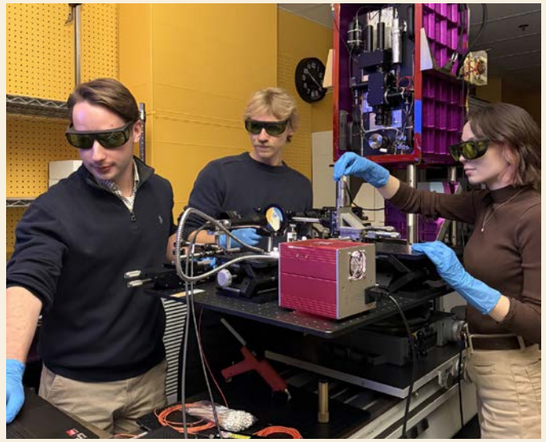
Jaiden Luna is a junior at California Polytechnic State University, where he majors in Aerospace Engineering. Supporting Sun Wong's FireTech project, Luna played an important role as the team integrated their instrument and prepared it for airborne flights, setting up the lab and preparing build instructions. Luna also helped design and assemble optical ground support equipment. "My experience has undoubtedly been a huge influence on my career trajectory and has been a great introduction to the field of optical engineering," said Luna. In his free time, Jaiden plays chess and tennis, and serves as the mechanical architect for his school's PolyIRover team.

Laela Bitahey

Laela Bitahey is a sophomore engineering student at the University of Arizona. As a member of Meredith Kupinski's IIP team, Bitahey measured the transmission of float-zone silicon wafers and the noise equivalent differential temperature for the instrument's LWIR camera and fore optics. "As an Indigenous woman, this experience has reaffirmed my purpose to continue my education in science and engineering. I hope to carry the knowledge I've gained and give back to my community and continue to increase indigenous representation in STEM fields," said Bitahey. Bitahey enjoys volunteering, playing the cello, and collecting vinyl records.

GORDON Team

Three students have been working with Matthew McGill on his IIP project, Global Orbital Research with a Diurnal Observing Network (GORDON), an effort to produce a cost-efficient, spaceborne lidar instrument. The students, Jackson Begolka, Grant Finneman, and Sabrina Vlk, are all PhD students at the University of Iowa. Together, they have been developing interface control documents, testing lidar instrument components, developing data collection and processing algorithms, and testing the polarization sensitive camera. "IIP projects can provide excellent training opportunities for students, allowing them to gain hands-on experience while learning how projects are conceived and implemented," said McGill. "It should be a point of pride at ESTO that these three students will all use GORDON as part of their PhD theses."



Luke Jacobs

Luke Jacobs is a third-year PhD candidate in Electrical and Computer Engineering at the University of Illinois, Urbana-Champaign. Supporting Elahe Soltanaghai's FireTech project, Jacobs helped develop electromagnetic models of forests, as well as signal processing algorithms for drone-mounted radar systems dedicated to observing wildfires. "This experience helped me develop my research skills and understanding of radar physics," Jacobs said. In his free time, Jacobs enjoys reading and hosting Bible study with his friends.